 # Smart Security System

An Arduino-based security system that uses an HC-SR04 ultrasonic sensor to detect intruders and triggers audio-visual alarms. Built with an Arduino Uno R3 and a prototype expansion shield.

How It Works

The system operates across three states:

- **DISARMED** -- System is idle. Green LED on, screen shows "SYSTEM DISARMED". No sensor monitoring.

- **ARMED** -- Press the button to arm. Red LED on, screen shows live distance readings from the ultrasonic sensor updated every 200ms. If an object is detected within 50cm continuously for 3 seconds, the alarm triggers.

- **ALARM** -- Buzzer produces a 300ms ON / 200ms OFF siren pattern. Red LED blinks at 250ms intervals. TFT display flashes "INTRUDER ALERT" between red and yellow every 500ms. The alarm auto-silences after 30 seconds and returns to ARMED, or you can press the button to return to DISARMED immediately.

Hardware

| Component | Model / Type |

|-----------|-------------|

| Microcontroller | Arduino Uno R3 (ATmega328P) |

| Expansion | Prototype shield (I/O ports, 26p connector, prototyping area, I2C) |

| Distance sensor | HC-SR04 ultrasonic (4-pin) |

| Display | 1.8" ST7735S TFT LCD (128x160, SPI) |

| Buzzer | Active buzzer |

| Indicators | Red LED + Green LED (330 ohm resistors) |

| Input | Push button (INPUT_PULLUP) |

Pin Assignments

| Shield Pin | Component | Signal |

|---|---|---|

| D2 | Push button | Leg A |

| D3 | Red LED | Anode (+) |

| D4 | Green LED | Anode (+) |

| D5 | HC-SR04 | TRIG |

| D6 | HC-SR04 | ECHO |

| D7 | Active buzzer | + |

| D8 | TFT | RES |

| D9 | TFT | DC |

| D10 | TFT | CS |

| D11 | TFT | SDA (SPI MOSI) |

| D13 | TFT | SCL (SPI SCK) |

| 5V | TFT VCC, TFT BL, HC-SR04 VCC | Power |

| GND | All components | Ground |

Software Setup

Libraries required

Install via Arduino IDE Tools > Manage Libraries:

- Adafruit GFX Library

- Adafruit ST7735 and ST7789 Library

Upload

1. Open `SmartSecuritySystem.ino` in Arduino IDE
2. Select Tools > Board > Arduino Uno
3. Select Tools > Port (your Uno's COM port)
4. Click Upload
5. Open Serial Monitor at 9600 baud

Serial Monitor Output

The system logs events to Serial Monitor at 9600 baud:

Tag	When	Example
`[BTN]`	Button press confirmed	`[BTN] Button pressed`
`[STATE]`	State transition	`[STATE] DISARMED -> ARMED`
`[DIST]`	Distance reading (every 500ms while ARMED)	`[DIST] 38.2 cm`
`[GRACE]`	Grace period counting (every loop while active)	`[GRACE] Elapsed: 1200 ms`

Key Design Details

- All timing uses `millis()`-based non-blocking logic -- no `delay()` calls in the main loop
- Button input is debounced with a 50ms window
- Grace period uses a start-time model: any non-detection or invalid reading resets the 3-second window immediately
- Alarm auto-silences after 30 seconds and returns to ARMED
- TFT updates use `fillRect()` for partial screen refresh to avoid flicker

- The distance formula uses the standard HC-SR04 conversion: `distance = (pulseDuration / 2.0) / 29.1` (cm)

Project Structure

```
SmartSecuritySystem/
  SmartSecuritySystem.ino   # Main sketch (all code in one file)
  demo.md                   # Serial Monitor demo walkthrough
  wiring.md                 # Complete pin-by-pin wiring guide
  README.md                 # This file
```

License

This project is open source. Feel free to use, modify, and share.